Weak-Net Dini Convergence Characterizes Normal Cones

A proposed solution of Open Problem 6.2 in Glück

June 21, 2026

Abstract

Let X be a real ordered Banach space with closed pointed cone X_+. We prove that X_+ is normal if and only if every increasing weakly convergent net in X is norm convergent. In fact, it is enough to assume this only for norm-bounded nets. More precisely, if X_+ is non-normal, then there are $u \in X_+$ and a norm-bounded increasing net (y_i) in the order interval $[0, u]$ such that $y_i \rightharpoonup u$ while $\|y_i - u\| \to 1$. This gives an affirmative answer to Open Problem 6.2 of Glück and also settles the bounded-net variant mentioned there.

We also audit the supplied partial construction for the explicit non-normal cone on $\ell^1(\mathbb{N}_0)$. That construction is correct; it is a concrete special case of the general phenomenon proved here.

1 Problem and result

Throughout, an *ordered Banach space* means a real Banach space X endowed with a closed pointed cone X_+. The induced order is $x \leq y$ if $y - x \in X_+$. Recall that X_+ is *normal* if there is $M \geq 0$ such that

$$0 \leq x \leq y \quad \Longrightarrow \quad \|x\| \leq M\|y\|.$$

Glück asked whether normality follows from the property that every increasing weakly convergent net is norm convergent [2, Open Problem 6.2]. He also noted that the answer might conceivably change if one assumes, in addition, that the net is norm bounded. The following theorem answers both questions.

Theorem 1.1 (Weak-net Dini characterization)**.** *Let X be an ordered Banach space. The following assertions are equivalent.*

(i) *The cone X_+ is normal.*

(ii) *Every increasing weakly convergent net in X converges in norm.*

(iii) *Every norm-bounded increasing weakly convergent net in X converges in norm.*

If X_+ is non-normal, the counterexample to (iii) *can be chosen in the following strong form: there exist $u \in X_+$ and an increasing net $(y_i) \subseteq [0, u]$ such that*

$$y_i \rightharpoonup u, \qquad \sup_i \|y_i\| < \infty, \qquad \|y_i - u\| \longrightarrow 1.$$

The implication (i)⇒(ii) is the classical ordered-space version of Dini's theorem; see, for example, Schaefer–Wolff [4, Section V.4.3] and Glück [2, Corollary 4.2]. The content here is the converse, including the boundedness conclusion.

2 Audit of the supplied ℓ^1 construction

Consider $X = \ell^1(\mathbb{N}_0)$ with the cone

$$X_+ = \left\{ x \in \ell^1(\mathbb{N}_0) : x_0 \geq \sum_{k=1}^{\infty} \frac{|x_k|}{2^k} \right\}. \tag{1}$$

The source paper records that this cone is closed, generating, and non-normal, and asks specifically whether it nevertheless satisfies the net version of Dini's theorem [2, Example 6.1 and Open Problem 6.2].

Proposition 2.1. *The supplied partial proof constructs a valid increasing weakly convergent net in* (1) *which is not norm convergent.*

Proof. Write a vector as (a, u) with $u \in \ell^1(\mathbb{N})$, and set

$$q(u) = \sum_{k \geq 1} 2^{-k} |u_k|.$$

Then

$$(a, u) \leq (b, v) \quad \iff \quad b - a \geq q(v - u).$$

The supplied argument uses the directed set of pairs (n, F), where F is a finite subset of $\ell^\infty(\mathbb{N})$, and chooses $u_{n,F} \in \ell^1(\mathbb{N})$ satisfying

$$\|u_{n,F}\|_1 = 1, \qquad \operatorname{supp} u_{n,F} \subseteq \{k \geq n\}, \qquad \varphi(u_{n,F}) = 0 \quad (\varphi \in F).$$

Such a vector exists because a homogeneous system with $|F|$ equations and more than $|F|$ unknowns has a nonzero solution. Its support condition gives $q(u_{n,F}) \leq 2^{-n}$.

The order on the index set is defined so that a strict increase from (n, F) to (m, G) forces $m \geq n + 2$ and $F \subseteq G$. For

$$a_n = 4 - 2^{1-n}, \qquad y_{n,F} = (a_n, u_{n,F}),$$

the key estimate is

$$q(u_{m,G} - u_{n,F}) \leq 2^{-n} + 2^{-m} \leq 2(2^{-n} - 2^{-m}) = a_m - a_n,$$

where the middle inequality follows from $m \geq n + 2$. Thus the net is increasing. Given any $\varphi \in \ell^\infty(\mathbb{N})$, every sufficiently large finite-set component contains φ, so $\varphi(u_{n,F}) = 0$ eventually. Since $a_n \to 4$ along the directed set, $y_{n,F} \rightharpoonup (4, 0)$. Finally,

$$\|y_{n,F} - (4, 0)\|_1 = 1 + 2^{1-n},$$

so norm convergence fails. Moreover, $0 \leq y_{n,F} \leq 4e_0$, so this particular counterexample is both order bounded and norm bounded. All uses of the directed order, including transitivity, directedness, and eventual containment of a prescribed functional, are valid. $\square$

Thus the partial result completely settles the concrete ℓ^1 question. The remaining task is to remove the special weighted-tail geometry. The next two sections do so.

3 A principal order ideal and its finite-codimensional slices

Assume for this section that $u \in X_+$. Put

$$A_u = [-u, u] = \{x \in X : -u \leq x \leq u\}$$

and define the principal order ideal

$$E_u = \bigcup_{r>0} rA_u = \{x \in X : \text{for some } r > 0,\ -ru \leq x \leq ru\}.$$

The Minkowski functional of A_u is

$$p_u(x) = \inf\{r > 0 : x \in rA_u\}, \qquad x \in E_u.$$

The set A_u is absolutely convex and norm closed.

Lemma 3.1 (The order-unit norm and the inclusion map)**.** *The function p_u is a norm on E_u, its closed unit ball is exactly A_u, and the canonical inclusion*

$$J_u : (E_u, p_u) \longrightarrow (X, \|\cdot\|), \qquad J_u x = x,$$

has closed graph. If A_u is norm unbounded, then J_u is unbounded.

Proof. The only nonstandard point in the Minkowski-functional argument is positive definiteness. If $p_u(x) = 0$, then $x \in \varepsilon A_u$ for every $\varepsilon > 0$, hence

$$-\varepsilon u \leq x \leq \varepsilon u \qquad (\varepsilon > 0).$$

Letting $\varepsilon \downarrow 0$ and using norm closedness of X_+ gives $x \in X_+$ and $-x \in X_+$. Pointedness yields $x = 0$.

Certainly $A_u \subseteq \{p_u \leq 1\}$. Conversely, if $p_u(x) \leq 1$, then $x \in (1 + 1/n)A_u$ for every n, so $x/(1 + 1/n) \in A_u$. Norm closedness of A_u gives $x \in A_u$. Thus A_u is the closed p_u-unit ball.

To prove that J_u is closed, suppose that $x_n \to x$ in p_u and $x_n \to y$ in the norm of X. For each $\varepsilon > 0$, eventually $x_n - x \in \varepsilon A_u$. Since εA_u is norm closed, $y - x \in \varepsilon A_u$. This holds for every $\varepsilon > 0$, and the positive-definiteness argument above gives $y = x$. Finally, if A_u is norm unbounded, then the image under J_u of the p_u-unit ball is norm unbounded, so J_u is unbounded. ☐

The following lemma is the central device. It says that the unboundedness of an order interval cannot be concentrated in finitely many weak directions.

Lemma 3.2 (Finite-codimensional slice lemma)**.** *Suppose that $A_u = [-u, u]$ is norm unbounded. For every finite set $F \subseteq X'$ the set*

$$A_u \cap F^\perp, \qquad F^\perp := \bigcap_{f \in F} \ker f,$$

is norm unbounded.

Proof. Let

$$M = E_u \cap F^\perp.$$

Assume for a contradiction that $A_u \cap M$ is norm bounded, say by K. Since A_u is the p_u-unit ball, the restriction $J_u|_M$ is bounded:

$$\|m\| \leq K p_u(m) \qquad (m \in M). \tag{2}$$

We first show that M is closed in (E_u, p_u). Let $m_n \in M$ and $m_n \to m$ in p_u. By (2), (m_n) is norm Cauchy, so it converges in X to some y. Lemma 3.1 gives $y = m$. Since every $f \in F$ is norm continuous, $f(m) = \lim_n f(m_n) = 0$; hence $m \in M$.

The algebraic codimension of M in E_u is finite. Because M is p_u-closed, there is a finite-dimensional subspace $N \subseteq E_u$ such that

$$E_u = M \oplus N$$

topologically. Equivalently, there is $c > 0$ such that the unique decomposition $x = m + n$ satisfies

$$p_u(m) + p_u(n) \leq c\, p_u(x).$$

The inclusion J_u is bounded on M by (2), and it is bounded on the finite-dimensional space N. Therefore J_u is bounded on all of E_u, contradicting Lemma 3.1. □

Corollary 3.3. *Under the assumptions of Lemma 3.2, for every finite $F \subseteq X'$ and every $t > 0$ there is $x \in A_u \cap F^{\perp}$ with $\|x\| = t$.*

Proof. Choose $a \in A_u \cap F^{\perp}$ with $\|a\| \geq t$ and put $x = (t/\|a\|)a$. The set $A_u \cap F^{\perp}$ is balanced. □

4 Proof of the characterization theorem

Proof of Theorem 1.1. (i)⇒(ii). We include the standard Dini argument for completeness. Let $(x_i)_{i \in I}$ be increasing and suppose that $x_i \rightharpoonup x$. For each fixed i, the weak closedness of X_+ gives $x_i \leq x$.

Normality of X_+ implies that the dual cone X'_+ is generating. By the uniform decomposition theorem, there is $M \geq 1$ such that every $f \in X'$ can be written as $f = g - h$, where $g, h \in X'_+$ and

$$\|g\|, \|h\| \leq M\|f\|.$$

Consequently, with

$$K = X'_+ \cap \{f \in X' : \|f\| \leq 1\},$$

we have the norming estimate

$$\sup_{f \in K} f(v) \geq M^{-1}\|v\| \qquad (v \in X_+). \tag{3}$$

Indeed, for $v \neq 0$ choose $f \in X'$ with $\|f\| = 1$ and $f(v) = \|v\|$, decompose $f = g - h$, and use $\|v\| = g(v) - h(v) \leq g(v)$.

The set K is weak-star compact. Given $\varepsilon > 0$, define

$$U_i = \{f \in K : f(x - x_i) < \varepsilon/M\}.$$

The sets U_i are weak-star open in K and cover K, since for each positive f the nonnegative scalar net $f(x - x_i)$ tends to zero. Compactness gives indices $i_1, \ldots, i_n$ with $K = U_{i_1} \cup \cdots \cup U_{i_n}$. Choose i_0 dominating all i_k. Then monotonicity gives $K = U_{i_0}$, and (3) yields $\|x - x_i\| < \varepsilon$ for all $i \geq i_0$. Hence $x_i \to x$ in norm.

The implication (ii)⇒(iii) is immediate.

Contrapositive of (iii)⇒(i). Assume that X_+ is non-normal. A standard characterization of normal cones says that some order interval $[0, u]$, with $u \in X_+$, is norm unbounded; see [1, Theorem 2.40]. Hence $A_u = [-u, u]$ is norm unbounded.

Set

$$t_n = 4^n(1 + \|u\|) \qquad (n \in \mathbb{N}).$$

Let D consist of all pairs $i = (n, F)$, where $n \in \mathbb{N}$ and $F \subseteq X'$ is finite. Define

$$(n, F) \preceq (m, G) \quad \iff \quad (n, F) = (m, G) \text{ or } (m \geq n + 1 \text{ and } F \subseteq G).$$

This is a directed order: two indices are dominated by an index whose finite-set component is their union and whose integer component is larger than both.

For every $i = (n, F)$, Corollary 3.3 gives $x_i \in A_u \cap F^{\perp}$ such that

$$\|x_i\| = t_n.$$

Define

$$z_i = \frac{2u + x_i}{t_n}.$$

Since $-u \leq x_i \leq u$,

$$\frac{u}{t_n} \leq z_i \leq \frac{3u}{t_n}. \tag{4}$$

In particular, $0 \leq z_i \leq u$, because $t_n \geq 4$. If $i = (n, F) \prec j = (m, G)$, then $t_m \geq 4t_n$, and

$$z_j \leq \frac{3u}{t_m} \leq \frac{u}{t_n} \leq z_i.$$

Thus (z_i) is decreasing.

Fix $f \in X'$. Once the finite-set component contains f, we have $f(x_i) = 0$. Therefore, eventually,

$$f(z_i) = \frac{2f(u)}{t_n} \longrightarrow 0.$$

Hence $z_i \rightharpoonup 0$. On the other hand, the reverse triangle inequality and $\|x_i\| = t_n$ give

$$1 - \frac{2\|u\|}{t_n} \leq \|z_i\| \leq 1 + \frac{2\|u\|}{t_n}. \tag{5}$$

The integer component tends to infinity along D, so $\|z_i\| \to 1$. In particular, (z_i) is norm bounded but not norm null.

Finally put

$$y_i = u - z_i.$$

By (4), $0 \leq y_i \leq u$; since (z_i) is decreasing, (y_i) is increasing. Moreover,

$$y_i \rightharpoonup u, \qquad \|y_i - u\| = \|z_i\| \longrightarrow 1.$$

The estimate (5) also gives

$$\sup_i \|y_i\| \leq \|u\| + 1 + \frac{2\|u\|}{t_1} < \infty.$$

Thus (iii) fails whenever the cone is non-normal, completing the proof. $\square$

Remark 4.1 (Why a net is essential). The finite-set component of the directed index set forces the chosen vector to annihilate each prescribed functional eventually. In a space with the Schur property, such as ℓ^1, no analogous weakly null normalized *sequence* exists. The construction uses the full weak neighborhood filter and therefore genuinely exploits nets.

5 The explicit cone on ℓ^1

Corollary 5.1. *For the cone* (1)*, there is a norm-bounded increasing net* $(y_i) \subseteq [0, e_0]$ *such that*

$$y_i \rightharpoonup e_0 \quad \textit{but} \quad \|y_i - e_0\|_1 \not\longrightarrow 0.$$

Proof. For each $k \geq 1$, the vector

$$w_k = \frac{1}{2}(e_0 + 2^k e_k)$$

satisfies $0 \leq w_k \leq e_0$, while $\|w_k\|_1 = \frac{1}{2} + 2^{k-1} \to \infty$. Thus $[0, e_0]$ is norm unbounded. Apply the counterexample construction in Theorem 1.1 with $u = e_0$. $\square$

The supplied construction is more explicit than this corollary: it gives closed formulas for the scalar buffer and uses weighted-tail vectors of ℓ^1-norm one. The general proof shows that this is not a peculiarity of that cone; every non-normal closed cone admits a bounded weak-net Dini failure.

6 Literature check and novelty status

The literature search was carried out through June 21, 2026. It included exact-phrase searches for Open Problem 6.2, searches for combinations of "weakly convergent increasing net", "normal cone", and "Dini", forward citation checks for Glück's article, and inspection of the principal older references on monotone nets and ordered-space Dini theorems.

Glück's paper was published in 2025 and explicitly leaves the question open [2]. McArthur's classical paper studies convergence of order-bounded monotone nets and regularity of cones [3], while Wong and Ng study the sequence version of a Dini property [5]. The standard references prove the forward implication from normality to norm convergence of weakly convergent monotone nets, but the searches did not locate the converse proved in Theorem 1.1. The only forward citation to Glück's article that was located was unrelated to this open problem.

Accordingly, the theorem above appears to be a full answer not already present in the sources located. This is a literature-search conclusion, not a claim that an unindexed or differently phrased prior proof cannot exist.

References

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